

Key points:

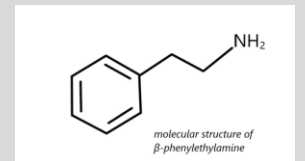
Drug classes

The mechanistic classes of the sympathomimetics are:

- **Direct-acting** drugs bind to and activate adrenergic receptors producing a response
- **Indirect-acting** drugs augment the activity of the catecholamine
 - Drugs that induce catecholamine release from storage vesicles → increase neurotransmitter availability
 - Drugs that inhibit reuptake into adrenergic neurons → increase synaptic concentrations of the catecholamine
 - Drugs that block degradation of catecholamines → increase catecholamine concentrations
- **Mixed-acting**
 - Drugs that directly stimulate adrenergic receptors AND induce release from storage vesicles

Structure-activity relationship → actions, uptake, and pharmacokinetic properties

- Most adrenergic drugs are derivatives of β -phenylethylamine (right figure).
- Substitutions on the benzene ring or on the ethylamine side chains produce a variety of compounds with varying abilities to differentiate between α and β receptors and to penetrate the CNS.
- Two important structural features of these drugs are (1) the number and location of OH substitutions on the benzene ring and (2) the nature of the substituent on the amino nitrogen.
- The structural characteristics determine potency at the receptors, uptake by norepinephrine transporter, and pharmacokinetic properties, such as ability to penetrate into the CNS, metabolism by COMT and MAO, and half-life.



Key points

Physiology-pharmacology relationship

- The sympathetic nervous system 1) increases heart rate and blood pressure, 2) dilates airways, 3) dilates pupils, 4) mobilizes energy stores, and 5) redirects blood flow to the muscles by constricting peripheral and mesenteric vessels and simultaneously dilating the skeletal muscle vasculature.
- Sympathomimetic drugs mimic these actions.
- The pharmacologic effects are dependent on the mechanisms of each drug – nonspecific vs. specific actions on receptors, direct vs. indirect actions.

Pharmacologic effects → therapeutic uses

- See the following slide.

Actions → adverse effects and cautions

- Adverse effects are predictable because of well-understood mechanisms of action and widespread distribution of adrenergic receptors. They mimic the actions of sympathetic nervous system stimulation leading to a wide range responses.
- Intensity of effects is dose-dependent. Higher doses lead to exaggerated responses.
- When administered together, sympathomimetics enhance the effects of other drugs with sympathomimetic effects potentially leading to hypertensive crisis.

Key points: Therapeutic uses

- Sympathomimetics are an important group of drugs used for cardiovascular, respiratory, and other conditions. They are classified into subgroups based on their spectrum of action: direct action by binding and activating adrenergic receptors (alpha, beta, and/or dopamine), indirect action by increasing synaptic levels of endogenous norepinephrine, which activates adrenergic receptors, or mixed-actions (both direct and indirect actions).
- Endogenous adrenergic receptor agonists epinephrine, norepinephrine, and dopamine are available as pharmacotherapeutics – vasopressors (induce vasoconstriction).
- Epinephrine is the drug of choice in acute anaphylaxis. It prevents or reverses airflow obstruction due to bronchoconstriction, prevents or reverses cardiovascular collapse, and decreases release of mediators from mast cells.
- Norepinephrine is the first-line agent for the initial treatment of shock (dangerously low hypotension and inadequate organ perfusion – life-threatening).
- Dopamine is an alternative for norepinephrine in selected patients with shock.
- Phenylephrine is a selective alpha-1 receptor agonist for treatment of patients with shock that have failed therapy with other vasopressors. It is also used topically for treatment of nasal and conjunctival congestion.
- Isoproterenol is rarely used.
- Dobutamine, a beta-1 receptor agonist, is administered for short-term use in decompensated heart failure for its inotropic effects.

Cardiovascular terminology in this presentation

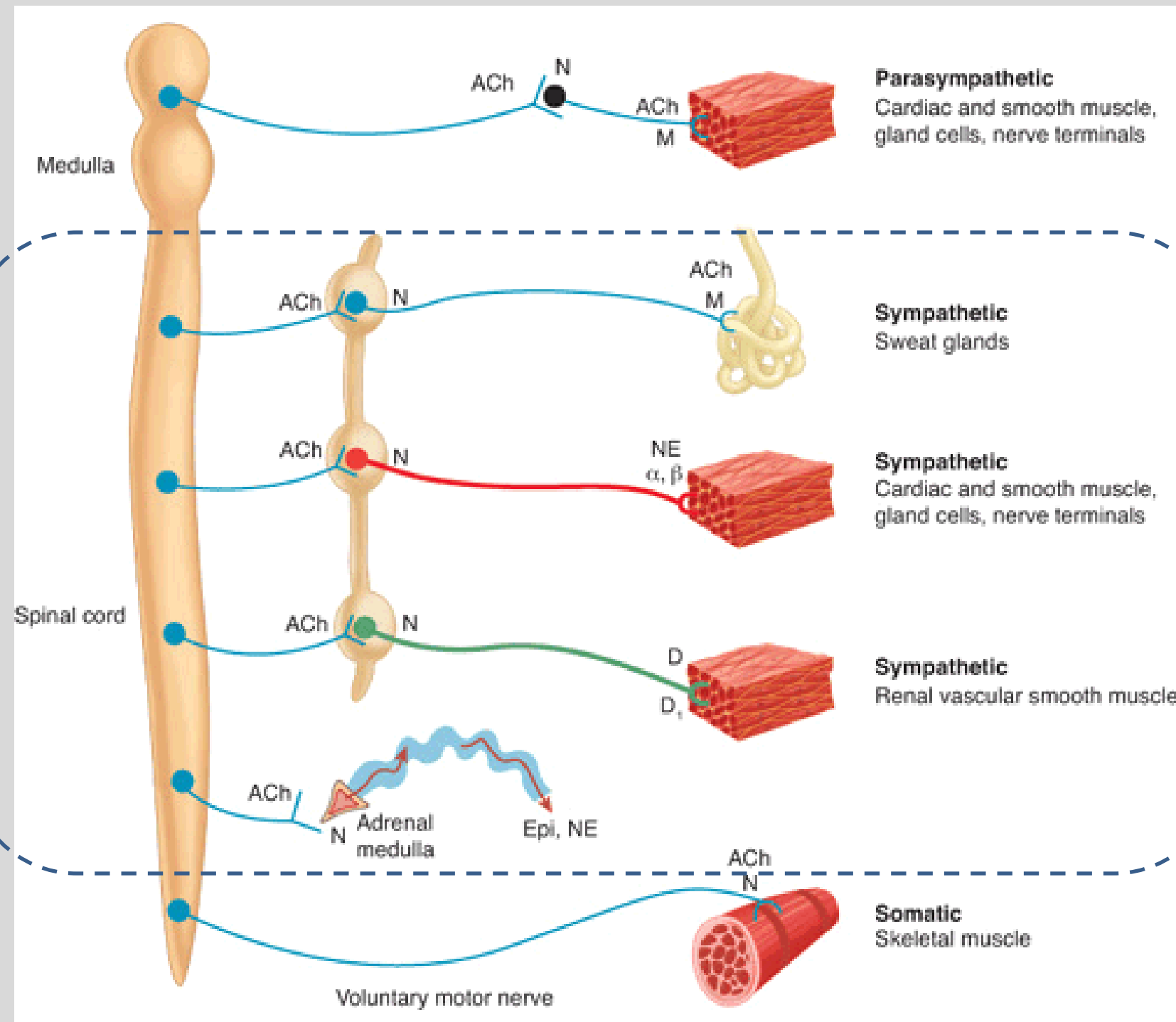
Heart rate (HR)	The number of times the heart beats per minute. Normal HR is 60-100 bpm.
Stroke volume (SV)	The amount of blood pumped out of the left ventricle in one heart beat.
Cardiac output (CO)	The total amount of blood the heart pumps per minute: $HR \times SV = CO$
Peripheral resistance Total peripheral resistance (TPR)	The resistance by blood vessels against the flow of blood. Total peripheral resistance is the amount of force the entire systemic vasculature (except the pulmonary circulation) exerts on blood flow. The main contributors are the smallest arteries and arterioles.
Mean arterial pressure (MAP)	The average arterial pressure throughout one cardiac cycle – systole and diastole – is influenced by cardiac output and total peripheral resistance.
Preload	The volume of blood in the ventricles just before they contract. Preload is influenced by the venous return to the heart and the compliance of the ventricles.
Afterload	The pressure (the load) the heart must overcome to push blood out of the ventricles into the systemic circulation. Afterload is primarily determined by the total peripheral resistance and the aortic pressure.

Cardiovascular terminology in this presentation

Systolic blood pressure (SBP)	BP increases to maximum pressure as left ventricular ejection (systole) of blood into aorta increases pressure and distends the vascular wall, which creates a tension wave along the artery wall. SBP is influenced by cardiomyocyte contraction, stroke volume, arterial compliance, and total peripheral resistance.
Diastolic blood pressure (DBP)	The minimum pressure in the arteries between heartbeats when ventricles relax and refill with blood (diastole) is influenced by total peripheral resistance, blood volume, and blood viscosity.
Pulse pressure	The difference between SBP and DBP ($SBP - DBP = \text{pulse pressure}$) (typically 40 to 60 mmHg in a young healthy adult)
Baroreceptors	Receptors located in the carotid sinuses and aortic arch are sensitive to changes in the stretch of arterial walls that occur with changes in mean arterial pressure to maintain blood pressure within normal levels.

Cardiovascular terminology in this presentation

Reflex responses to changes in BP	• When blood pressure rises, baroreceptors detect greater stretch in the arterial walls and send a signal via glossopharyngeal nerve (CN IX) to the brainstem, which increases vagal signaling (CN X), resulting in a lower heart rate (bradycardia) and vasodilation, which lowers blood pressure. • When the blood pressure decreases, baroreceptors send a signal to the brainstem, which increases sympathetic signaling resulting in increased heart rate (tachycardia) and vasoconstriction, raising blood pressure.
Bradycardia	Heart rate <60 bpm
Tachycardia	Heart rate >100 bpm (sinus tachycardia)
Renin (ree' nin)	An enzyme secreted from specialized cells in the kidney when blood pressure drops too low. Renin increases levels of angiotensin II, a powerful vasoconstrictor and stimulator of aldosterone secretion. Aldosterone causes reabsorption of sodium and water (RAAS – renin-angiotensin-aldosterone system). Increased sympathetic tone activates β1ARs in the kidneys, which induces renin release, activating the RAAS.



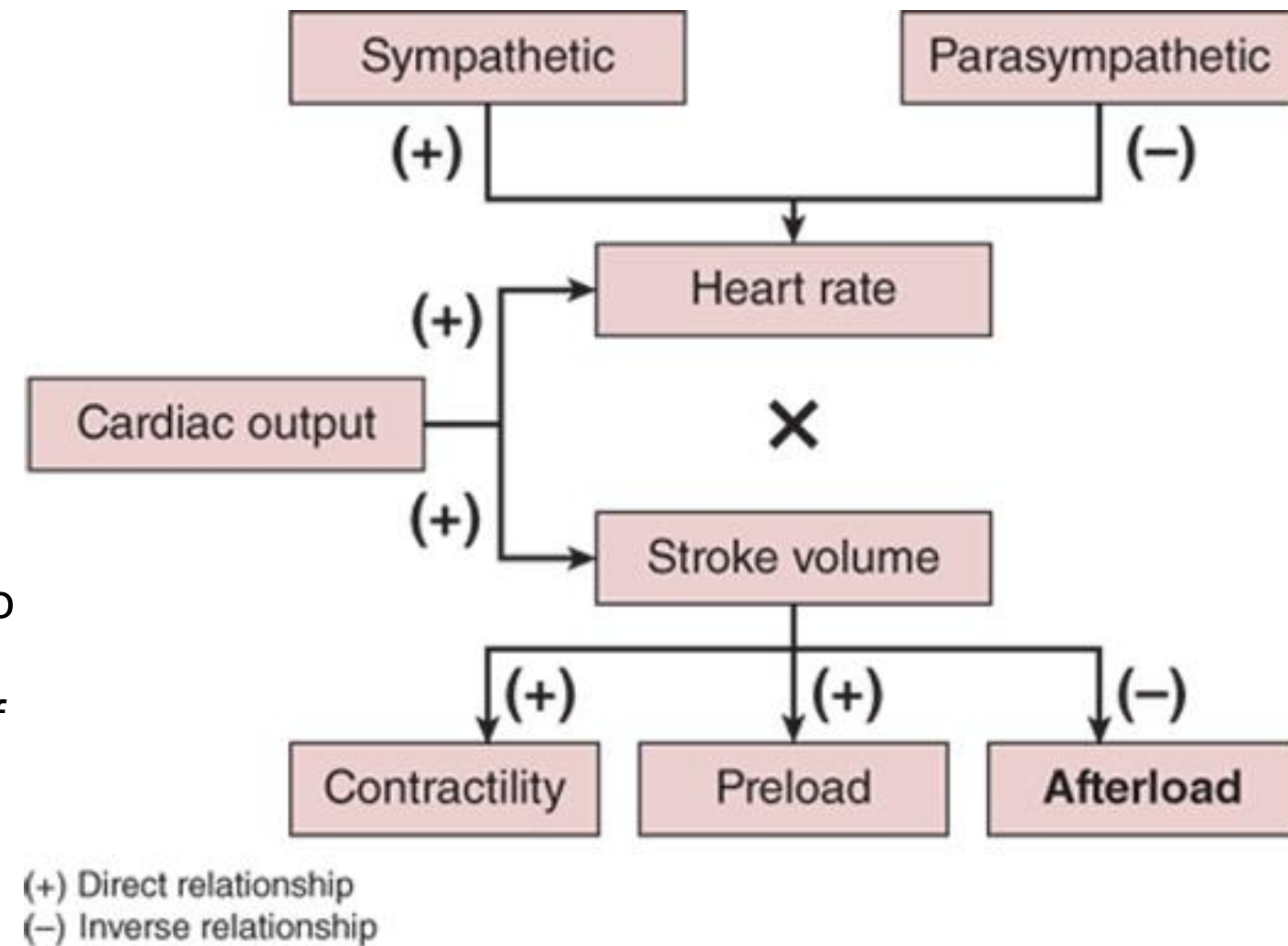
- The sympathetic nervous system controls cardiovascular effects and activities that increase arousal and energy expenditures and inhibit digestion.

Main factors regulating cardiac output.

- **Cardiac output (CO)** varies based on the body's requirements and depends on the heart rate (beats per minute) and the stroke volume (blood volume pumped per heartbeat):

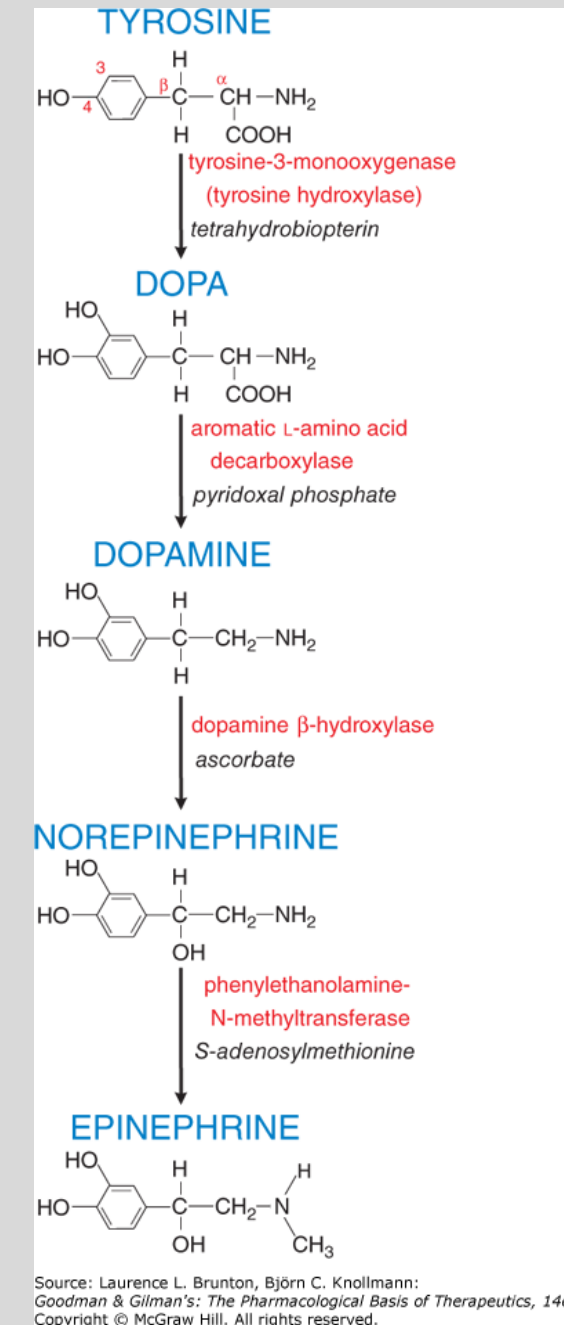
$$\text{CO} = \text{HR} \times \text{SV}$$

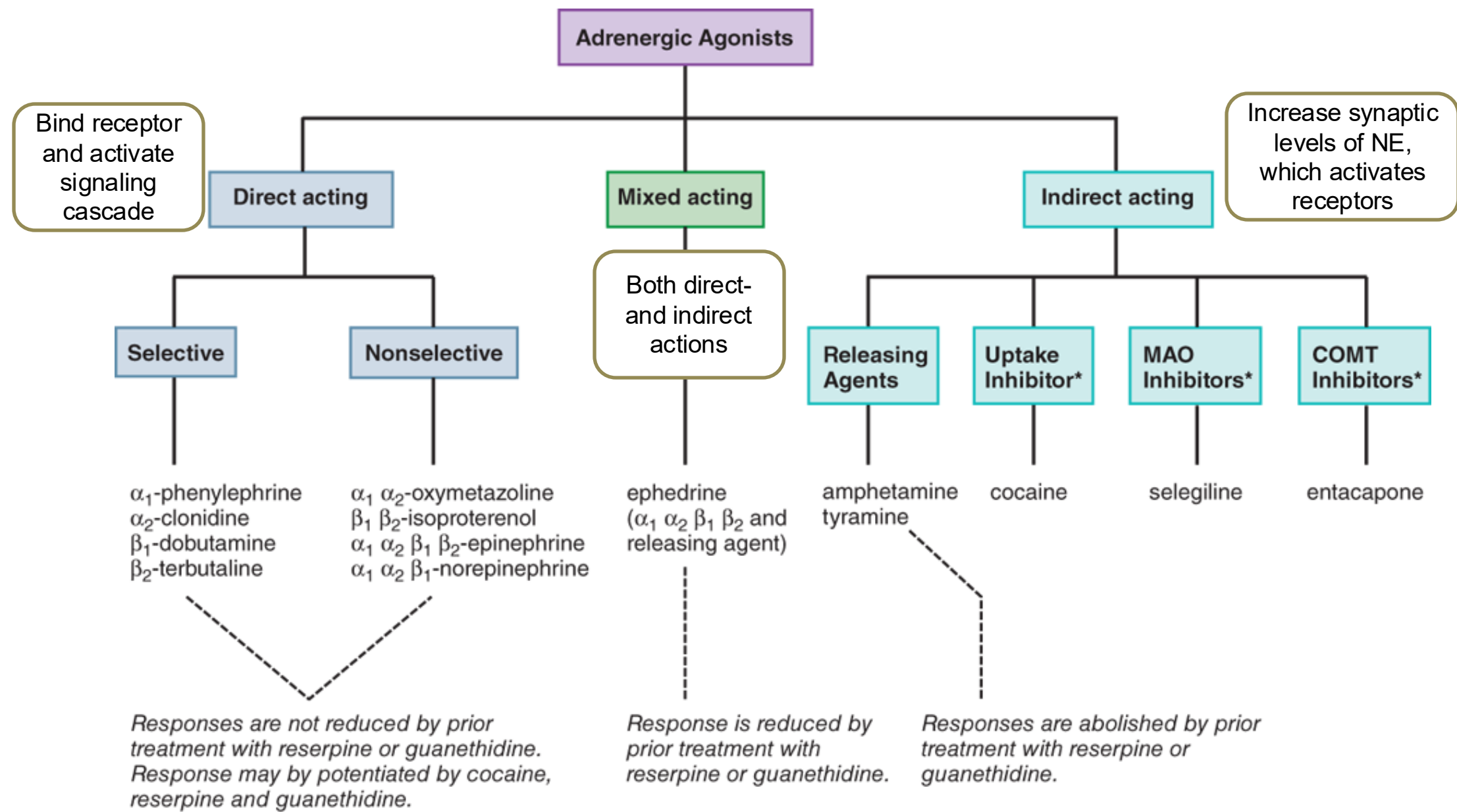
- **Heart rate (HR)** is determined by the balance between sympathetic and parasympathetic activation, normally, on the sinoatrial (SA) node of the heart.
- **Stroke volume (SV)** is the volume of blood ejected from each ventricle during the contraction in 1 heartbeat. It is equivalent to the difference between the volume of blood in the ventricle just before the contraction (end-diastolic volume) and the volume of blood left in the ventricle after the contraction (end-systolic volume).
- **Preload** is defined as the stretching of the myocardial muscle fibers just prior to a contraction or ventricular wall tension at the end of diastole. The more blood returning to the heart (within physiologic limits) as venous return, the more blood pumped from the heart as cardiac output.
- **Afterload** is defined as the ventricular wall stress during the ejection phase of systole or the resistance that the ventricle needs to overcome in order to eject its content – the amount of aortic pressure that the heart ejects blood against in order to empty the left ventricle. Afterload is determined by several factors, including wall stress, aortic pressure, and total peripheral resistance.



Source: Adel Elmoselhi:
Cardiology: An Integrated Approach
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Pharmacology of Sympathomimetics





Classification of adrenergic receptor agonists (sympathomimetic amines) or drugs that produce sympathomimetic effects. For each category, a prototypical drug is shown.

Endogenous Epinephrine's Physiologic Effects ("Fight or Flight" response)

Summarizing EPI's physiologic actions and effects provides context for pharmacologic EPI's clinical activity.

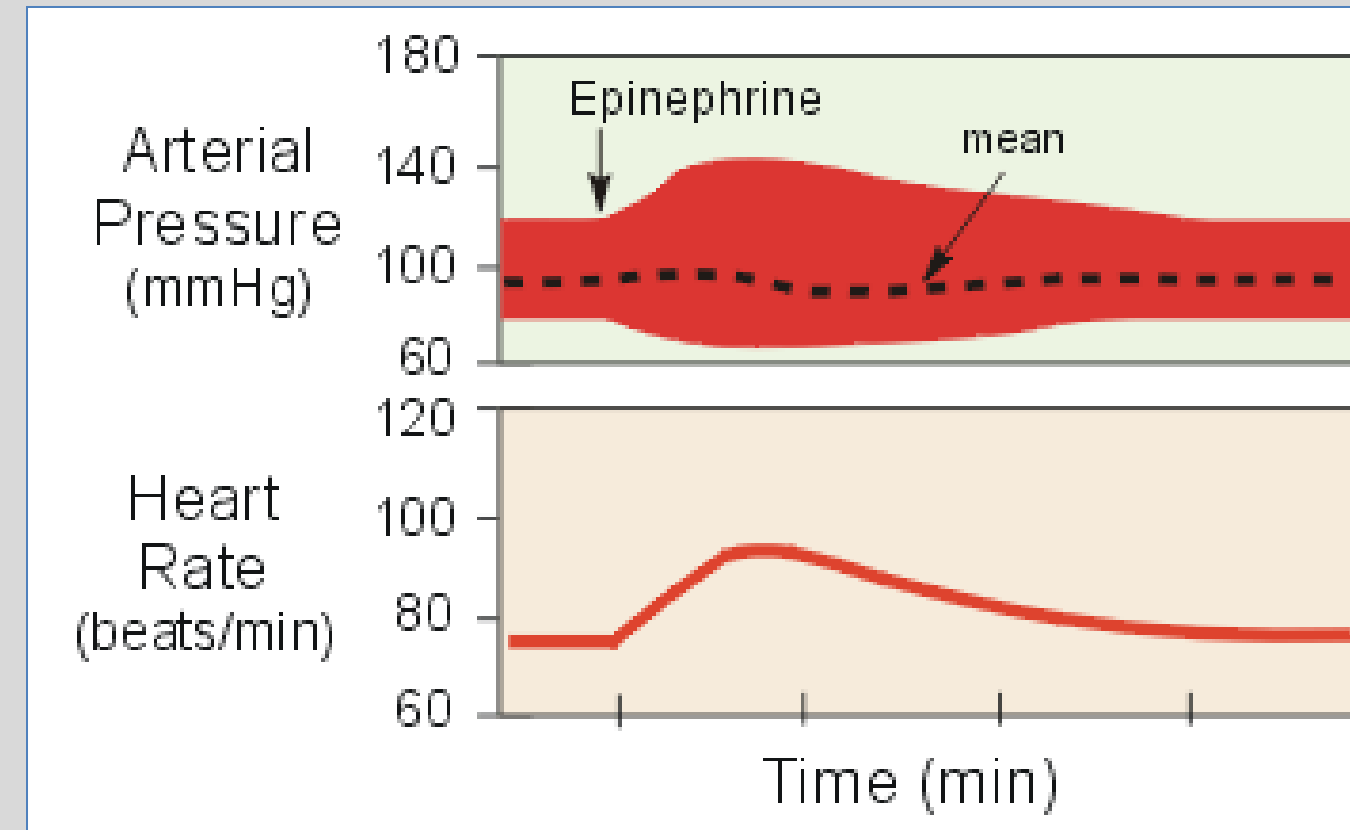
Cardiac (β_1 ARs)	• $\uparrow$ heart rate • $\uparrow$ stroke volume • $\uparrow$ cardiac output • $\uparrow$ coronary blood flow • proarrhythmic	Respiratory (β_2 ARs)	• Bronchodilation
Blood pressure (cardiac & vascular)	• $\uparrow$ systolic blood pressure • usually $\downarrow$ diastolic pressure • MAP only slight change • $\uparrow$ mean pulmonary pressure	Metabolic	• $\uparrow$ oxygen consumption • $\uparrow$ blood glucose • $\uparrow$ lactic acid • $\downarrow$ plasma K^+
Peripheral circulation (α_1 & β_2 ARs)	• $\downarrow$ or unchanged TPR • slight $\uparrow$ cerebral blood flow • $\uparrow$ skeletal muscle blood flow (β_2) • $\downarrow$ cutaneous blood flow (α_1) • $\downarrow$ splanchnic blood flow (α_1)	CNS	• $\uparrow$ respiration • $\uparrow$ subjective sensations
		Immune system	• Various immunomodulatory actions

Epinephrine's dose-dependent effects on blood pressure

1. Low dose epinephrine

- **β_2 -mediated** enhanced blood flow to skeletal muscles $\rightarrow$ $\downarrow$ peripheral resistance $\rightarrow$ $\downarrow$ diastolic blood pressure
- **β_1 -mediated** $\uparrow$ heart rate (HR) and $\uparrow$ force of contraction $\rightarrow$ $\uparrow$ stroke volume (SV), $\uparrow$ cardiac output, $\uparrow$ cardiac oxygen demand
- **Mean blood pressure** (MAP) usually not greatly elevated with low-dose EPI

Little change in MAP $\rightarrow$ compensatory baroreceptor reflexes do not appreciably antagonize direct cardiac actions.



<http://www.cvphysiology.com/Blood%20Pressure/BP018.htm>

EPI effects on BP at intermediate and rapid administration infusion rates

2. Intermediate rate of epinephrine infusion



greater degree of
 α -mediated vasoconstriction



compensatory reflexes
may be activated as MAP rises

3. Rapid infusion of epinephrine

i.e. high concentration (DANGEROUS)



predominantly α effects
(including skeletal muscle vasculature)



$\uparrow$ TPR + $\uparrow$ HR and $\uparrow$ contractility
 α -mediated vasoconstriction β -mediated cardiac effects



$\uparrow$ systolic and diastolic blood pressure



Reflex homeostatic adjustments

$\rightarrow \downarrow$ heart rate

Epinephrine Pharmacokinetics

A	Epipen IM autoinjector, subQ, IV, IO, intranasal spray, oral inhalation IV / IO route: Caution; Careful use by IV or IO route <i>only</i> if immediate, reliable effect is required (eg profound hypotension or cardiopulmonary arrest) - Not absorbed orally – rapidly metabolized in GI mucosa and liver IO: intraosseous
D	Catecholamines do not cross the blood-brain barrier.
M	Circulating epinephrine: MAO, COMT degradation in many tissues, particularly the liver
E	Metabolites and small amount of unchanged compound normally excreted in urine <i>Clinical correlate:</i> High levels of Epi, NE, and metabolites in blood and urine are indicative of pheochromocytoma.
onset	IM rapid; subQ ~5-10 minutes; intranasal spray ~1 minute IV immediate



Epinephrine Therapeutic Uses and Class Adverse Effects

Therapeutic uses:

- IM and intranasal spray (**without delay!**)
 - **Anaphylactic shock reversal**
- **Drug of Choice**
never contraindicated for anaphylaxis treatment
- Subcutaneously with local anesthetics
 - local vasoconstriction
- Oral inhalation via nebulizer
 - acute asthma bronchodilation
- IV for cardiovascular effects:
 - myocardial stimulation during cardiopulmonary resuscitation
 - alternative to NE in septic shock
- Topical hemostat

Adverse effects:

- Restlessness, apprehension, throbbing headache, tremor, palpitations
- Premature ventricular contractions
- Ventricular arrhythmias due to excessive stimulation
- Angina ($\uparrow O_2$ consumption)
- Cerebral hemorrhage
(large dose $\rightarrow$ sharp $\uparrow$ BP)

Cautious use:

Conditions when response to EPI could lead to severe hypertension or cardiac complications

Adverse effects:

Sympathetic stimulation

Extravasation at infusion site (leakage) $\rightarrow$ tissue necrosis

Check your knowledge

EPI is the drug of choice for what condition?

Why is it the drug of choice (actions and effects)?

Norepinephrine Effects and Therapeutic Uses

Potent vasopressor (α effects)

- Vasoconstriction increases total peripheral resistance $\rightarrow$ $\uparrow$ afterload
- Constriction of veins $\rightarrow$ $\uparrow$ venous return to the heart $\rightarrow$ $\uparrow$ preload and stroke volume

Cardiac effects (β_1 effects)

- $\uparrow$ cardiac force of contraction $\rightarrow$ modest $\uparrow$ cardiac output
- $\uparrow$ MAP $\rightarrow$ reflex parasympathetic effect $\rightarrow$ heart rate unchanged or decreased

THERAPEUTIC USES: Circulatory shock

Profound hypotension with low peripheral resistance

- Septic | Cardiogenic | Neurogenic

drug of choice for initial therapy

based on clinical experience of effectiveness

- *Selection of therapy depends on patient-specific factors.*
- *Provide adequate blood volume restoration.*

ADVERSE EFFECTS typical of the class:

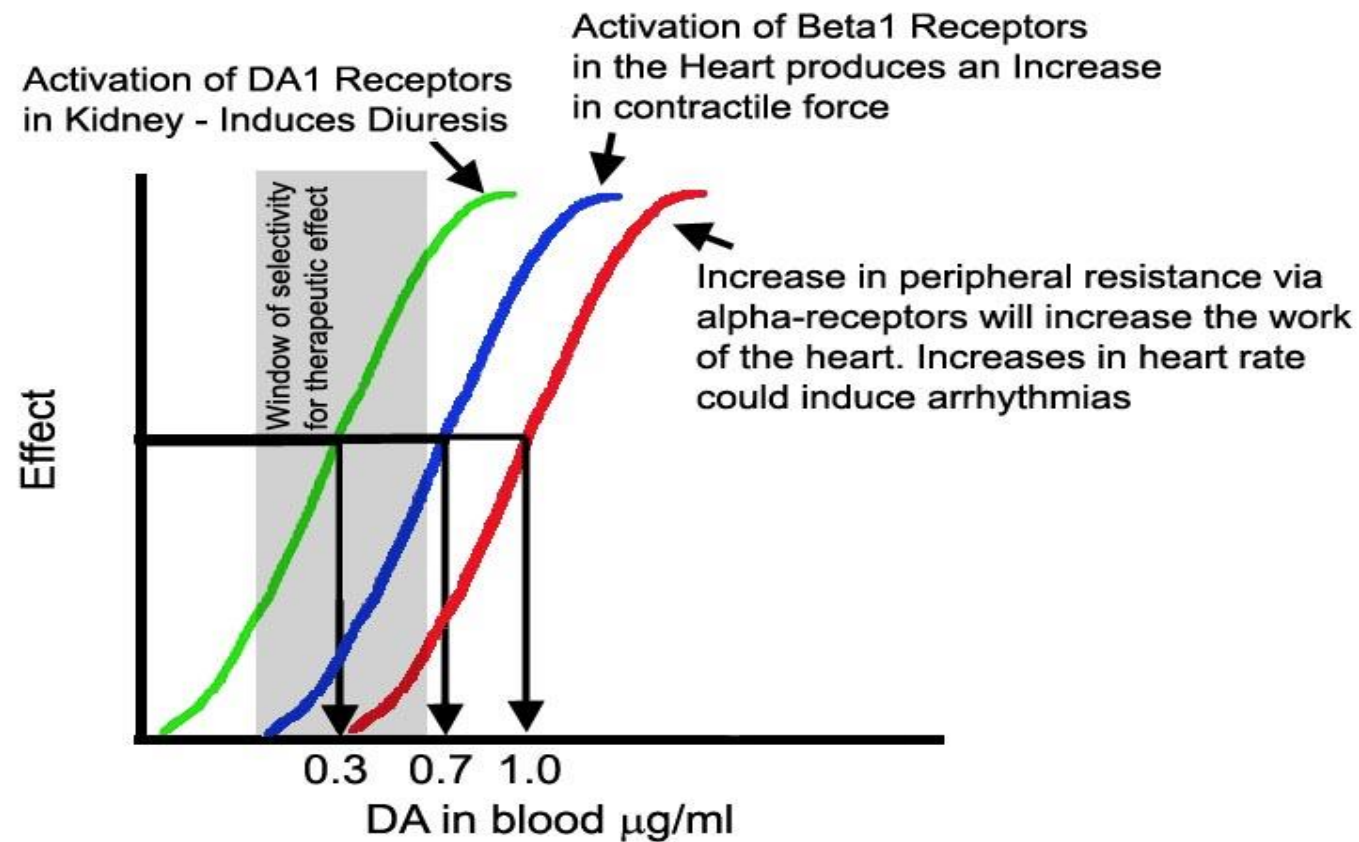
- Reduced blood flow to organs $\rightarrow$ $\downarrow$ organ perfusion and danger of organ failure
- Ventricular arrhythmia
- Digital ischemia $\rightarrow$ gangrene
- Extravasation at injection site $\rightarrow$ tissue necrosis, gangrene

Check your knowledge

NE is the drug of choice for what conditions?

Why is it the drug of choice (actions and effects)?

Dopamine's Dose-Dependent Hemodynamic Effects



Therapeutic use: alternative to NE for septic and cardiogenic shock

- Dopamine was associated with higher mortality and dysrhythmias were more common (subgroup analysis from a randomized trial)
- Dopamine does not appear to provide any renal protective advantage over routine therapy with IV diuretics.

Low dose (1-5 mcg/kg/min)

D₁ activation in renal and mesenteric vessels → **vasodilation**

- ↑ renal blood flow, ↑ GFR, and ↑ Na⁺ excretion
- ↓ peripheral vascular resistance

Intermediate dose (5-10 mcg/kg/min)

β₁ ARs activation → ↑ contractility (direct effect) + causes NE release from nerve terminals (indirect effect) → positive inotropic effect

- ↑ systolic blood pressure; diastolic pressure same or slightly increased
- Total peripheral resistance usually unchanged

High-dose (>10 mcg/kg/min)

α₁ ARs activation → general vascular constriction (pressor effect)

Isoproterenol

Relative affinities	Potent, nonselective β AR agonist
activity at β_1 AR, β_2 AR and β_3 AR	No affinity for α ARs
β_1 : ISO >> EPI > NE = DA	β_2 : ISO > EPI >>> NE > DA

- Cardiac: positive **chronotropic** and **inotropic** actions $\rightarrow$ $\uparrow$ cardiac output and $\uparrow$ O₂ consumption
- Nonvascular: β_2 -mediated relaxation of bronchial and GI smooth muscle

β_2 AR stimulation $\rightarrow$

- $\downarrow$ peripheral vascular resistance
- $\downarrow$ diastolic BP
- systolic BP may be unchanged or rise
- $\downarrow$ MAP

Limited Therapeutic Uses: Treatment of hypotension resulting from bradycardia

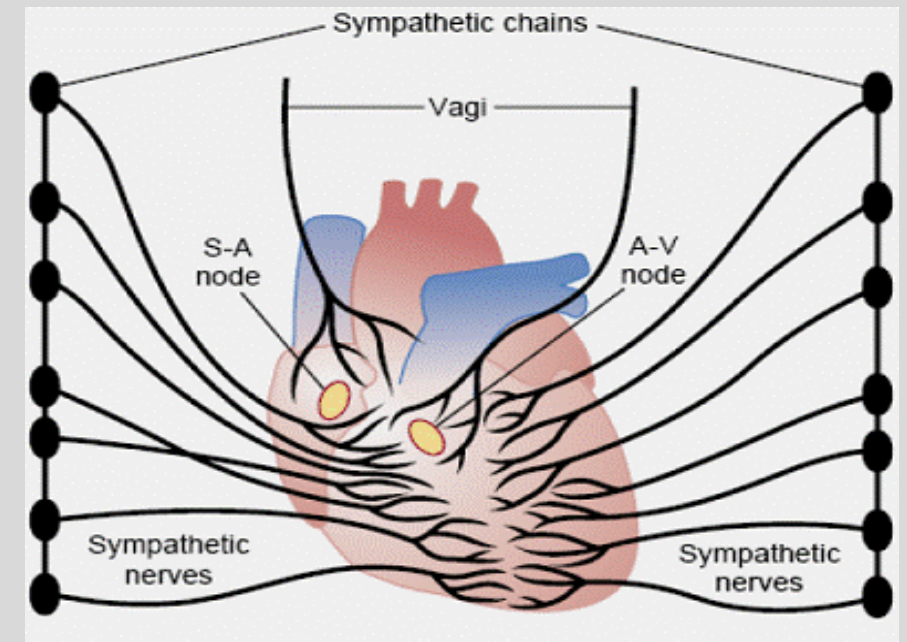
Adverse Effects

- Reflex tachycardia $\rightarrow$ Prominent chronotropic effect

Cardiac ischemia / arrhythmias can occur in patients with underlying coronary or cardiac disease.

Dobutamine: β Agonist **Net Effects**

- Dobutamine has predominantly β_1 receptor agonist action and a mild β_2 receptor agonism.
- It causes β_1 -mediated increased cardiac contractility and heart rate, resulting in increased cardiac output ($CO = HR \times \text{stroke volume}$).
- Its β_2 activating effects may be slightly greater than its α_1 stimulating effects resulting in some vasodilation.



🔑 An advantage of dobutamine use compared to isoproterenol is increased cardiac output usually with a relatively mild increase in heart rate due to less reflex tachycardia.

↑ cardiac output	↓ systemic vascular resistance (↑CO → reflex vasodilation + β_2 effect)
↓ central venous pressure and ↓ pulmonary wedge pressure (indicators of preload)	improves renal blood flow

At low doses, dobutamine relieves the signs and symptoms of congestive heart failure.

THERAPEUTIC USES

- Cardiac decompensation, due to cardiac surgery, CHF, acute MI
 - short-term treatment
- Dobutamine stress testing in patients unable to perform exercise stress testing (with echocardiogram)
- Long-term dobutamine use is associated with decreased survival.

CHF: congestive heart failure; MI: myocardial infarction

ADVERSE EFFECTS

- Tachyphylaxis:
24h-72h continuous use → agonist-induced β AR desensitization / receptor downregulation requiring dose adjustment
- Dose-dependent rise in systolic pressure and heart rate → hypertension, tachycardia, angina, arrhythmia
Hypertension, angina, arrhythmia, tachycardia related to the β_1 actions

Comparing the effects of NE, EPI, and isoproterenol

NE: α -mediated vasoconstriction $\rightarrow \uparrow \text{TPR} + \uparrow \text{SBP}$ and $\uparrow \text{DBP}$ and $\uparrow \text{MAP}$

- Potent vasoconstriction causes an increase in MAP $\rightarrow$ which causes a modest increase in heart rate (direct activation of β_1 in the SA node) $\rightarrow$ followed by reflex bradycardia in response to increased MAP $\rightarrow$ which results in a heart rate that is unchanged or mildly decreased.

EPI (physiologic concentrations):

β_1 activation $\rightarrow \uparrow$ rate and force of cardiac contraction $\rightarrow \uparrow \text{SBP}$

- β_2 -mediated vasodilation in skeletal muscle vascular beds $\rightarrow \downarrow \text{TPR} \rightarrow \downarrow \text{DBP}$
- MAP is modestly increased due to α -mediated vasoconstriction in other vascular beds so is *unlikely to trigger the reflex effect* on heart rate.
- Activation of β_1 cardiac receptors $\rightarrow \uparrow \text{HR}$ and $\uparrow$ inotropy

Isoproterenol:

- β_2 -mediated vasodilation in skeletal muscle vascular beds *in the absence of α -mediated vasoconstriction*, along with vasodilation in renal and mesenteric vasculature $\rightarrow \downarrow \text{DBP} \rightarrow$ significant $\downarrow \text{TPR} \rightarrow \downarrow \text{MAP} \rightarrow$ reflex $\uparrow \text{HR}$
- Reflex $\uparrow \text{HR}$ plus + β_1 -induced $\uparrow \text{HR} \rightarrow$ tachycardia, which can lead to dysrhythmias

- The sympathomimetics are grouped into subclasses related to their direct effects on adrenergic receptors, the indirect-acting sympathomimetics, mixed-acting agents.
- The structural variations of the drugs determine the relative affinities for adrenergic receptors, catecholamine transporters, and to degradation by monoamine oxidase and catechol-O-methyltransferase.
- The pharmacological properties of both the direct- and indirect-acting drugs relate to the classification, distribution, and mechanisms of the α and β receptors.
- Depending on their receptor specificities, the drugs mimic the physiologic effects characteristic of the catecholamines, including increasing heart rate and force of cardiac contraction, modifying systemic peripheral resistance, metabolic and endocrine actions, CNS actions on respiratory stimulation, wakefulness and psychomotor activity, and appetite reduction.
- Sympathomimetic vasopressors and inotropes are mainstays in the management of shock and heart failure.
- Adverse effects are extensions of their effects on α and β receptors: excessive vasoconstriction, cardiac arrhythmias, myocardial infarction, hemorrhagic stroke, pulmonary edema, and more.

Check your knowledge

EPI is the drug of choice for what condition?

- Anaphylaxis / anaphylactic shock (hypotension due to widespread vasodilation and vascular leakage leading to reduced flow to organs and organ failure)

Why is it the drug of choice (actions and effects)?

- Anaphylactic shock is deadly. EPI works fast and prevents progression of symptoms.
- It causes bronchodilation (β_2 action) which opens airways
- It causes vasoconstriction (α_1 , α_2) and increases the force of cardiac contraction(β_1) which increase blood pressure.
- Vasoconstriction also reduces allergic angioedema (with throat swelling) and hives
- Suppresses release of histamine and other mediators from mast cells

Check your knowledge

NE is the drug of choice for what conditions?

- Circulatory shock: profound hypotension with low peripheral resistance
- septic shock, cardiogenic shock, neurogenic shock
- Drug selection is based on patient-specific factors.

Why is it the drug of choice (actions and effects)?

- Clinical experience of NE's effectiveness.
- Potent vasoconstriction to increase systemic vascular resistance and raise blood pressure
- Mild increase in cardiac output compared to phenylephrine
- Lower risk of arrhythmias compared to dopamine and dobutamine
- NOTE: Vasopressors are components of a broader treatment plan.

Check your knowledge

What are the dose-dependent effects of intravenous dopamine infusion on blood pressure, chronotropy, inotropy, and peripheral vascular resistance?

- Low dose: **D₁** activation in renal and mesenteric vessels → vasodilation → ↑renal blood flow, ↑GFR, and ↑Na⁺ excretion and ↓ peripheral vascular resistance
- Intermediate dose: **β₁ ARs** activation → ↑contractility (direct effect) + causes NE release from nerve terminals (indirect effect) → positive inotropic effect → ↑systolic blood pressure; diastolic pressure same or slightly increased and total peripheral resistance is usually unchanged
- High dose: **α₁ ARs** activation → general vascular constriction (pressor effect)

What are the effects of intravenous phenylephrine infusion on blood pressure, chronotropy, inotropy, and peripheral vascular resistance?

- Phenylephrine cause generalized vasoconstriction → ↑peripheral resistance, ↑SBP, and ↑DBP with marked reflex bradycardia and stroke volume may decrease (slowed HR + lack of positive inotropic effect)

What are the clinical indications for dopamine and phenylephrine?

- Both are alternatives for NE in the treatment of shock.
- Phenylephrine also has topical applications for reducing congestion by inducing local vasoconstriction.